

Fawaz

SUMMARY

A collection of custom PCB projects designed in KiCad, showcasing schematic design, PCB layout, routing, ground plane implementation, and design rule verification.

PROJECTS

RGB Mood Light PCB | Arduino Nano ESP32 based

- 6× WS2812 RGB LEDs
- USB-C power input
- Push button control
- 2-layer PCB
- DRC: 0 Errors, 0 Warnings

ESP32 OLED Weather / System Display PCB | Arduino Nano ESP32

- OLED display interface
- Bosch BME280 environmental sensor
- I²C communication
- Dual-layer ground planes
- USB-C powered
- DRC: 0 Errors, 0 Warnings

TECHNICAL SKILLS

KiCad

Arduino Nano ESP32

PCB Design Workflow

ERC and DRC Validation

Schematic Capture and Layout Design

Schematic Design

Component Placement

PCB Routing

Ground Plane Design

Via Stitching

USB-C Integration

I²C Bus Design

Design Debugging and Verification